Convex optimization for machine learning – TD3

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1 Exercises

Exercise 1. (Proximal gradient for logistic regression)

We consider a classification problem defined by observations $(\mathbf{x}_i, y_i)_{1\leq i\leq n}$ where for all i, $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{-1, 1\}$. The coordinates of $\mathbf{x}_i$ are the explanatory variables of the model. We note $x_{i,j}$ the i-th observation of the j-th explanatory variable, so that $\mathbf{x}_i = (x_{i,1}, \cdots, x_{i,p})$.

We propose the following linear model for the generation of the data. Each observation is supposed to be independent and there exists a vector $\mathbf{w} \in \mathbb{R}^p$ and $w_0 \in \mathbb{R}$ such that for all i, $(y_i, \mathbf{x}_i)$ is a realization of the random variable $(Y, \mathbf{X})$ whose law satisfies

$$\mathbb{P}_{w_0,\mathbf{w}}(Y = 1|\mathbf{X}) = \frac{\exp(\mathbf{X}^\top \mathbf{w} + w_0)}{1 + \exp(\mathbf{X}^\top \mathbf{w} + w_0)} \,.$$

1. In this model, what are the explanatory variables? The observed outcomes? The parameters? Can you think of a real-life situation to which you could apply this model?

2. We set $\sigma := x \mapsto (1 + \exp(-x))^{-1}$; this function is known as the *logistic function*.

   Show that $\forall i \in [\![1, n]\!]$, $\mathbb{P}_{w_0,\mathbf{w}}(Y_i = y_i|\mathbf{x}_i) = \dfrac{1}{1 + \exp(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0))} = \sigma\big(y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)$.

3. Show that the maximum likelihood estimator is

$$(\widehat{w_0}, \widehat{\mathbf{w}}) = \arg\min_{w_0,\mathbf{w}} \sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big)$$

4. Denote $f(w_0, \mathbf{w}) = \sum_{i=1}^n \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big)$. Compute $\nabla f(w_0, \mathbf{w})$.

5. Compute the proximal operator of $\mathbf{x} \mapsto \frac{\lambda}{2}\|\mathbf{x}\|^2$.

6. Write the proximal gradient method for the logistic regression problem with ridge regularizer

$$(\widehat{w_0}^\lambda, \widehat{\mathbf{w}}^\lambda) = \arg\min_{w_0,\mathbf{w}} \sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big) + \frac{\lambda}{2}\|\mathbf{w}\|^2.$$

7. Compute the Hessian matrix and write Newton's method for the same problem.

Exercise 2. (LASSO)

We consider the problem

$$\min_{\mathbf{x}\in\mathbb{R}^n} \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda\|\mathbf{x}\|_1 \,.$$

1. Prove that the solution is $\{0\}$ for large λ.

2. For an arbitrary λ, provide the expression of the proximal gradient algorithm, using the step size $\gamma_k := 1/L$.

3. Assume that the initial point is at distance D from a minimizer. How many iterations are needed (at most) to achieve an ε-minimizer?

2 Solutions

Exercise 1 1. The explanatory variables are the $\mathbf{x}_i$, and the observations are the y_i. The parameters are $\mathbf{w}$ and w_0. This model is usually applied to solve binary prediction problems. For example, to predict if a patient will have diabetes ($y = 1$) or not ($y = -1$) according to explanatory variables $\mathbf{x} = (x^1, \cdots, x^p)$, like his weight, his age, his job, *etc.* It could also be used to predict if someone will vote for a certain party or not, according to various social and economic indicators. This model is closer from classification methods than true regression.

2.
- Assume first $y_i = 1$. By multiplying the numerator and the denominator of $\mathbb{P}_{w_0,\mathbf{w}}(Y = y_i|\mathbf{X})$ by $\exp(-(\mathbf{X}^\top\mathbf{w} + w_0))$, we get:

$$\mathbb{P}_{w_0,\mathbf{w}}(Y = y_i|\mathbf{X}) = \frac{\exp(\mathbf{X}^\top\mathbf{w} + w_0)}{1 + \exp(\mathbf{X}^\top\mathbf{w} + w_0)} = \frac{1}{1 + \exp(-(\mathbf{X}^\top\mathbf{w} + w_0))} = \frac{1}{1 + \exp(-y_i(\mathbf{X}^\top\mathbf{w} + w_0))}.$$

- Assume now that $y_i = -1$. Then:

$$\mathbb{P}_{w_0,\mathbf{w}}(Y = y_i|\mathbf{X}) = 1 - \mathbb{P}_{w_0,\mathbf{w}}(Y = 1|\mathbf{X}) = \frac{1}{1 + \exp(\mathbf{X}^\top\mathbf{w} + w_0)} = \frac{1}{1 + \exp(-y_i(\mathbf{X}^\top\mathbf{w} + w_0))}.$$

3. Remember that the likelihood L of a discrete probability distribution μ_θ in the observations $\mathbf{y}_1, \cdots, \mathbf{y}_n$ and with parameter θ is the probability of observing all these observations at the same time, assuming they are independent:

$$L(\theta, \mathbf{y}_1, \cdots, \mathbf{y}_n) = \mathbb{P}(Y_1 = y_1, \cdots, Y_n = y_n) = \prod_{i=1}^{n} \mathbb{P}(Y_1 = y_i) = \prod_{i=1}^{n} \mu_\theta(\{y_i\}),$$

where the Y_i are i.i.d. random vectors with distribution μ_θ. In particular they have the same distribution as Y_1. Furthermore the maximizers of a positive function f are the same as those of $\log \circ f$, so we will focus on the log-likelihood ℓ rather than on the likelihood itself, since it is easier to deal with. Here $\theta = (w_0, \mathbf{w})$ and we want to find $(\widehat{\mathbf{w}}, \widehat{w}_0)$ which are the likeliest parameters of the model, owing to the available data (observations + explanatory variables), that is, those which maximize:

$$\ell(\widehat{\mathbf{w}}, \widehat{w}_0, \mathbf{x}_1, \cdots, \mathbf{x}_n) = -\sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top\widehat{\mathbf{w}} + \widehat{w}_0)\big)\Big).$$

Because of the minus sign, this means finding the minimizers of:

$$f : (w_0, \mathbf{w}) \mapsto \sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top\mathbf{w} + w_0)\big)\Big),$$

and so the maximum likelihood estimator(s) of $(w_0, \mathbf{w})$ are the points $(\widehat{\mathbf{w}}, \widehat{w}_0)$ which minimize this function.

4. It suffices to compute the partial derivatives with respect to the $p + 1$ variables $w_0, w_1, \cdots, w_p$. We find:

$$\nabla f(w_0, \mathbf{w}) = \begin{pmatrix} -\sum_{i=1}^{n} y_i \sigma\big(-y_i(\mathbf{x}_i^\top\mathbf{w} + w_0)\big) \\ -\sum_{i=1}^{n} x_{i1} y_i \sigma\big(-y_i(\mathbf{x}_i^\top\mathbf{w} + w_0)\big) \\ \vdots \\ -\sum_{i=1}^{n} x_{ip} y_i \sigma\big(-y_i(\mathbf{x}_i^\top\mathbf{w} + w_0)\big) \end{pmatrix},$$

5. By definition, computing the proximal operator of $g = \mathbf{x} \mapsto \frac{\lambda}{2}\|\mathbf{x}\|^2$ means to find the minimizer[1] of:

$$h(\mathbf{x}) := \frac{\lambda}{2}\|\mathbf{x}\|^2 + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2 = \frac{\lambda+1}{2}\|\mathbf{x}\|^2 - \langle \mathbf{x}, \mathbf{y}\rangle + \frac{1}{2}\|\mathbf{y}\|^2.$$

This function is convex (as a sum of such functions) and its gradient is:

$$\nabla h(\mathbf{x}) = (\lambda+1)\mathbf{x} - \mathbf{y}.$$

As a result, its unique minimizer, *i.e.* the proximal operator of g, is

$$\mathrm{prox}_g(\mathbf{y}) = \frac{1}{\lambda+1}\mathbf{y}.$$

6. By definition, the recurrence relation given by the proximal gradient method for the logistic regression problem with given step sequence $(\gamma_k)_k$ is:

$$\begin{aligned}(w_0^{k+1}, \mathbf{w}^{k+1}) &= \mathrm{prox}_{\gamma_k g}\big((w_0^k, \mathbf{w}^k) - \gamma_k \nabla f(w_0^k, \mathbf{w}^k)\big) \\ &= \frac{1}{\lambda+1}\big((w_0^k, \mathbf{w}^k) - \gamma_k \nabla f(w_0^k, \mathbf{w}^k)\big).\end{aligned}$$

This result is consistent with the definition of the ridge regularization: when $\lambda = 0$, the penalization term vanishes and we find the usual gradient descent algorithm. And when $\lambda = +\infty$, the only value of $(\widehat{w_0}, \widehat{\mathbf{w}})$ which solves the logistic regression problem with ridge regularizer is $(\widehat{w_0}, \widehat{\mathbf{w}}) = (\mathbf{0}_{\mathbb{R}^p}, 0)$.

7. We must compute the cross partial derivatives $\partial_{i,j} f$ of the $\mathcal{C}^2$-class function $f : \mathbb{R}^{p+1} \to \mathbb{R}$. We set:

$$z_k := y_k(\mathbf{x}_k^\top \mathbf{w} + w_0).$$

Remember that the y_k are numbers equal to ± 1, so $y_k^2 = 1$. Also notice that $\sigma'(x) = e^{-x}\sigma(x)$, so that:

$$\frac{\mathrm{d}}{\mathrm{d}x}\sigma(-x) = -e^x \sigma(-x)^2.$$

- When $i = j = 1$, we differentiate the first term of the gradient with respect to the first variable. We find:

$$\partial_{1,1} f(w_0, \mathbf{w}) = \sum_{k=1}^{n} e^{z_k}\sigma(-z_k)^2.$$

[1] Which is unique, since g satisfies the hypotheses of the exercise 1 of the previous tutorial: g is convex and lower-semicontinuous with a nonempty domain.

- When $i \in [\![2, p+1]\!]$ and $j = 1$, we differentiate the i-th term of the gradient with respect to the first variable. We find:

$$\forall i \in [\![2, p+1]\!],\ \partial_{i,1} f(\mathbf{x}) = \sum_{k=1}^{n} x_{k,i} e^{z_k} \sigma(-z_k)^2.$$

- For $i, j \in [\![2, p+1]\!]$, we find:

$$\forall i, j \in [\![2, p+1]\!],\ \partial_{i,j} f(\mathbf{x}) = \sum_{k=1}^{n} x_{k,i} x_{k,j} e^{z_k} \sigma(-z_k)^2$$

Once the Hessian matrix is known, we must compute for each k the vector $(\nabla^2 f(\mathbf{x}_k))^{-1} \nabla f(\mathbf{x}_k)$. This means solving the system $\nabla^2 f(\mathbf{x}_k)\mathbf{x} = \nabla f(\mathbf{x}_k)$. In the present case there does not seem to exist a closed expression for the solution, so we have to resort to numerical solvers to find the solution $(\widehat{w_0}^{\lambda}, \widehat{\mathbf{w}}^{\lambda})$ of the logistic regression problem with ridge regularizer by using Newton's algorithm.

Exercise 2 1. We set:

$$f(\lambda, \mathbf{x}) = \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_1.$$

It is clear that:

$$\lim_{\lambda \to \infty} f(\lambda, \mathbf{x}) = \frac{1}{2}\|\mathbf{b}\|^2 + \iota_{\{\mathbf{0}\}}(\mathbf{x}) = \begin{cases} \frac{1}{2}\|\mathbf{b}\|^2 & \text{if } \mathbf{x} = \mathbf{0} \\ +\infty & \text{if } \mathbf{x} \neq \mathbf{0} \end{cases}.$$

But $f(\lambda, \mathbf{0}) = \frac{1}{2}\|\mathbf{b}\|_2^2$ for all $\lambda \geq 0$, so intuitively, as soon as λ is large enough, the minimum of $f(\lambda, \cdot)$ should be reached in $\mathbf{0}$, since $f(\lambda, \mathbf{x})$ tends to infinity when λ goes to infinity if $\mathbf{x}$ is not zero. The trouble is that for each fixed λ, the minimizer of $f(\lambda, \cdot)$ depends on λ, so it is not clear how it will behave when λ goes to infinity.

To understand that, we need a uniform control on the minimizer (as a function of λ). It will be provided by the norm of $A^\top b$. First notice that we work in finite dimension, so that all norms are equivalent. In particular, there exists $c > 0$ such that $\|\mathbf{x}\|_1 \geq c\|\mathbf{x}\|_2$. Thus:

$$\begin{aligned}
f(\lambda, \mathbf{x}) - f(\lambda, \mathbf{0}) &= \frac{1}{2}\|A\mathbf{x} - b\|_2^2 + \lambda\|\mathbf{x}\|_1 - \frac{1}{2}\|\mathbf{b}\|_2^2 \\
&= \frac{1}{2}\|A\mathbf{x}\|_2^2 - \langle A\mathbf{x}, \mathbf{b}\rangle + \lambda\|\mathbf{x}\|_1 \\
&\geq -\langle A\mathbf{x}, \mathbf{b}\rangle + \lambda\|\mathbf{x}\|_1 \\
&\geq -\langle A\mathbf{x}, \mathbf{b}\rangle + c\lambda\|\mathbf{x}\|_2 \\
&= \|\mathbf{x}\|_2 \Big(- \frac{\langle A\mathbf{x}, \mathbf{b}\rangle}{\|\mathbf{x}\|_2} + c\lambda \Big) \\
&= \|\mathbf{x}\|_2 \Big(- \frac{\langle \mathbf{x}, A^\top \mathbf{b}\rangle}{\|\mathbf{x}\|_2} + c\lambda \Big) \\
&\geq \|\mathbf{x}\|_2 \big(- \|A^\top \mathbf{b}\|_2 + c\lambda \big) \qquad (1)
\end{aligned}$$

and the term between parentheses is positive as soon as $\lambda \geq \|A^\top \mathbf{b}\|_2/c$. For such λ, $\mathbf{0}$ is always a minimizer of $f(\lambda, \cdot)$. It is the unique possible minimizer thanks to equation (1) (if $\mathbf{x}$ is a minimizer of $f(\lambda, \cdot)$ and is not null, then so is its norm and $f(\lambda, \mathbf{x}) - f(\lambda, \mathbf{0}) > 0$. Absurd.

2. Using the notations of the lecture notes, we take $f = \mathbf{x} \mapsto \|A\mathbf{x} - \mathbf{b}\|_2$ and $g = \mathbf{x} \mapsto \|\mathbf{x}\|_1$. It is clear that both functions are convex, and that f is differentiable, with a Lipschitz gradient:

$$\nabla f(\mathbf{x}) = A^\top (A\mathbf{x} - \mathbf{b})$$

(see the first lecture!). In that particular case, we can take the matrix norm of $A^\top A$ as a Lipschitz constant, since:

$$\begin{aligned} \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 &= \|A^\top A\mathbf{x} - A^\top A\mathbf{y}\|_2 \\ &\leq \sup_{\mathbf{z} \neq 0} \frac{\|A^\top A\mathbf{z}\|_2}{\|\mathbf{z}\|_2} \|\mathbf{x} - \mathbf{y}\|_2 \\ &= \|A^\top A\|_{2,2} \|\mathbf{x} - \mathbf{y}\|_2. \end{aligned}$$

Here the proximal operator is a bit more complicated to compute than usual, since we must solve the optimization problem:

$$\text{For all } \mathbf{y} \in \mathbb{R}^d, \text{ find } \operatorname*{argmin}_{\mathbf{x} \in \mathbb{R}^d} \Big\{ \lambda \|\mathbf{x}\|_1 + \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \Big\}.$$

Notice that the proximal operator will be a function from $\mathbb{R}^n$ to $\mathbb{R}^n$. Notice also that:

$$\lambda \|\mathbf{x}\|_1 + \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 = \sum_{i=1}^n \big(\lambda |x_i\| + \frac{1}{2} (x_i - y_i)^2 \big) = \sum_{i=1}^n f_i(\lambda, x_i),$$

with obvious notations. This means that we have to minimize a sum of functions with separate variables. In that case, it is obvious we just have to minimize each function separately, *i.e.* the solution $\mathbf{x} = (x_1, \ldots, x_n)$ will have coordinates x_i which minimize f_i for each i. So it is enough to compute the proximal operator of $x \mapsto |x|$. This is not hard, but the computation itself is a bit tedious because of the numerous cases to distinguish... Set $g : x \mapsto \lambda|x| + (x - y)^2/2$. We know that x minimizes g iff $0 \in \partial g(x)$. Since:

$$\partial g(x) = \begin{cases} \operatorname{sign}(x)\lambda + x - y & \text{if } x \neq 0 \\ [-\lambda, \lambda] - y & \text{if } x = 0. \end{cases}$$

we must distinguish two cases (I gladly acknowledge that it is not obvious at all just by looking at that expression of the subgradient!):

- **If** $|y| > \lambda$: then 0 cannot belong to $\partial g(0)$, so the minimizer of g must be some $x \neq 0$. If $x \in \mathbb{R}_+^*$, we have that

$$0 \in \partial g(x) \iff x = y - \lambda,$$

which is possible only if $y > \lambda$. Otherwise $y < -\lambda$, and we find for $x \in \mathbb{R}_-^*$:

$$0 \in \partial g(x) \iff x = y + \lambda,$$

so, by reuniting those two cases:

$$x = y + \operatorname{sign}(y)\lambda.$$

- **If** $|y| \leq \lambda$, then $0 \in [-\lambda, \lambda] - y = \partial g(x)$, and 0 is a minimizer of g.

Once more we can reunite all previous cases in one expression, just by factorising by $\operatorname{sign}(y)$:

$$x = \operatorname{sign}(y)\big(|y| - \lambda\big)_+,$$

where $y_+ = \max(y, 0)$ is the *positive part* of y. This function is called *soft thresholding operator* (in one dimension). More generally, the proximal operator of $\mathbf{x} \mapsto \lambda\|\mathbf{x}\|_1$ is just:

$$\operatorname{prox}_{\lambda\|\cdot\|_1}(\mathbf{y}) = \Big(\operatorname{sign}(y_1)\big(|y_1| - \lambda\big)_+, \ldots, \operatorname{sign}(y_n)\big(|y_n| - \lambda\big)_+\Big).$$

Knowing this, it is an easy matter to write the proximal gradient algorithm for $f(\lambda, \cdot)$ (be careful: it is necessary to replace λ by λ/L in the previous expression!).

3. Using the course, we know that:

$$f(\lambda, \mathbf{x}_k) - f(\lambda, \mathbf{x}^*) \leq \frac{L\|\mathbf{x}_k - \mathbf{x}^*\|_2^2}{2k} = \frac{LD^2}{2k},$$

where $L = \|A\|_{2,2}$. By definition, an ε-minimizer of $f(\lambda, \cdot)$ is a point $\mathbf{x}$ such that $f(\lambda, \mathbf{x})$ is at distance at most ε from the minimum $f(\lambda, \mathbf{x}^*)$. So we just need to find k such that:

$$\frac{LD^2}{2k} \leq \varepsilon.$$

With the bound provided by the course, we find that we need at least $k = LD^2/(2\varepsilon)+1$ iterations to be sure that $\mathbf{x}_k$ is an ε-minimizer for the LASSO problem.

Remarque 1.
- *This is a sufficient condition, not a necessary one: we may have already an ε-minimizer with less iterations, hence the need to get bounds as sharp as possible to avoid redundant iterations!*
- *It does not mean that $\mathbf{x}_k$ is actually close from $\mathbf{x}^*$! To determine that, stronger assumptions may be required (for example, strong convexity (see your lecture notes)).*